

■ SAFETY FEATURES

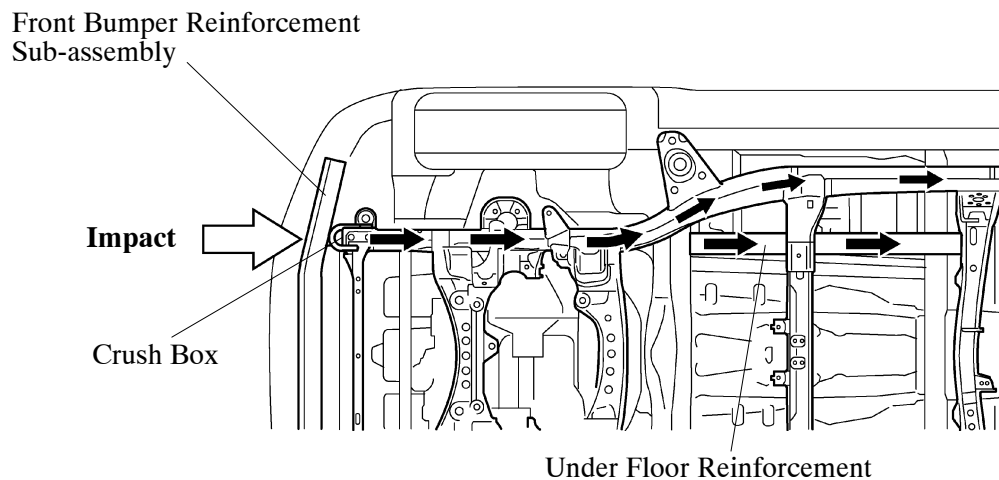
5. General

The impact absorbing structure of the new Land Cruise (Station Wagon) minimizes cabin deformation by effectively helping to absorb the impact energy in the event of a front, side or rear collision. This provides high-performance occupant protection.

6. Impact Absorbing Structure for Front Collision

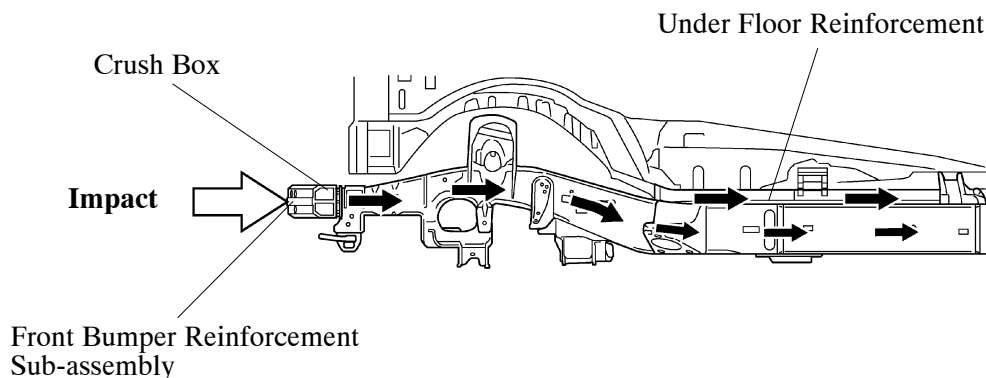
High strength sheet steel is used in the reinforcements located under the floor and on the rocker panels, and a new structure that can effectively provide the axial-compression load to the frame is used. This is to absorb the collision energy efficiently and to disperse the load. As a result, cabin deformation will be minimized. (See BO-2 and 3.)

- A large front bumper reinforcement is used to efficiently dissipate the impact energy into the frame side rails.
- Crush boxes are provided at the front ends of the frame side rails. These crush boxes reduce the impact that acts on the side rails and minimize body deformation during a minor collision.
- In order to disperse the impact load which is caused by a frontal offset collision, the frame structure has been designed to minimize the frame buckling and transfer collision energy more linearly. In addition, high strength sheet steel is used in the reinforcements under the floor. As a result, a more efficient dispersal of the collision impact load has been made possible by controlling frame distortion mode through a combination of the body and frame during a major collision.



Bottom View

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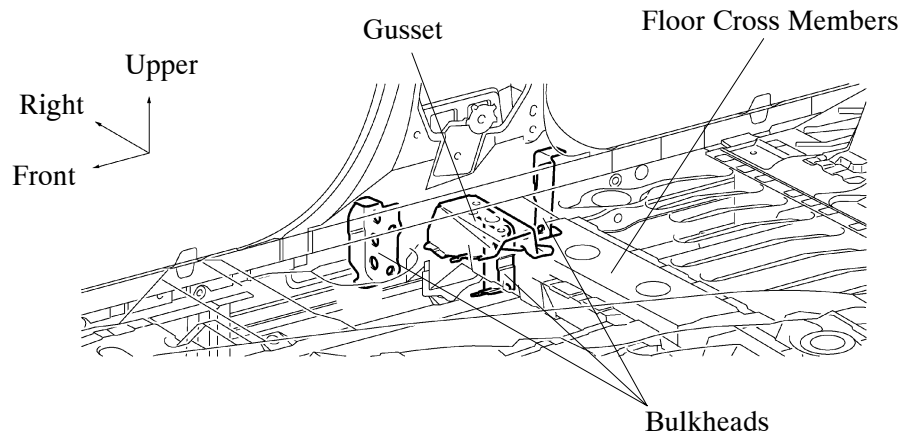
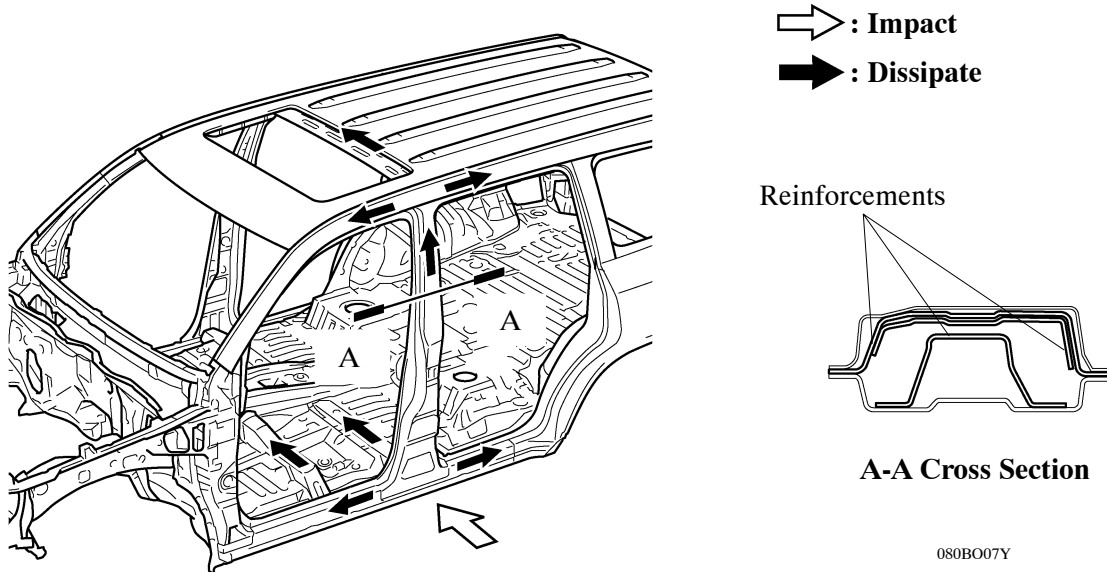
Side View

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7. Impact Absorbing Structure for Side Collision

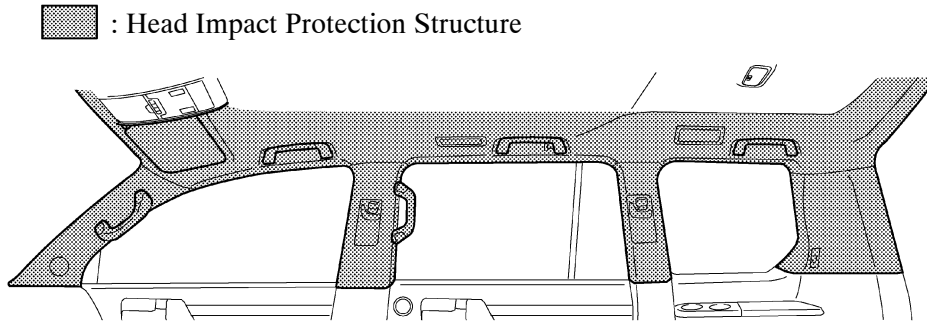
The body construction has been optimized to realize an effective transmission of the collision impact to the peripheral parts, such as from the center pillar to the roof reinforcements, rocker reinforcements, and floor cross members. This ensures the integrity of the cabin space and dramatically reduces body deformation.

- High strength sheet steel is used in the center pillar reinforcements, thus realizing a compact, lightweight, and highly rigid center pillar construction (A-A cross section).
- Bulkheads and gussets are placed at optimum locations within the rockers. As a result, the load from a center pillar can be transmitted to a floor cross member, absorbing the impact energy.



Rocker Member Section

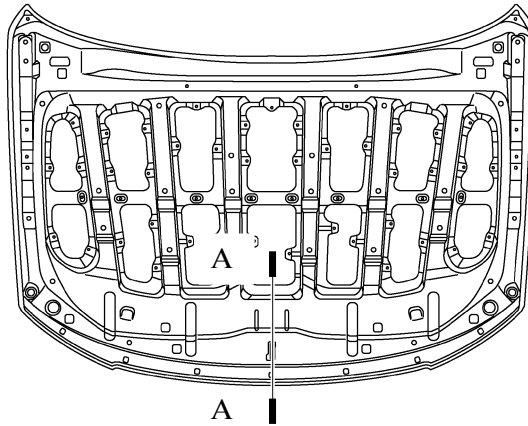
- A head impact protection structure is used. With this type of construction, if the occupant's head hits against the roof side rail or pillar due to a collision, the inner panels of the roof side rail and pillar collapse to help reduce the impact.



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8. Lessening Pedestrian Head Injury

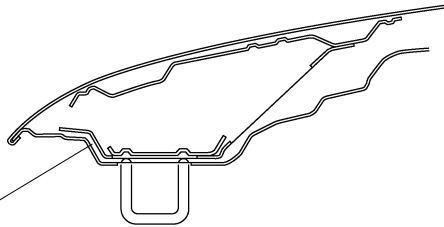
- In order to enhance the pedestrian protection performance, the hood has been designed with a longitudinal frame structure which brings uniform hood rigidity and efficiently absorbs the impact.
- An impact absorption shape has been provided at the front of the hood. If someone short such as a child hits the front, the impact will be absorbed more efficiently, and head injuries will be minimized.



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Engine Hood

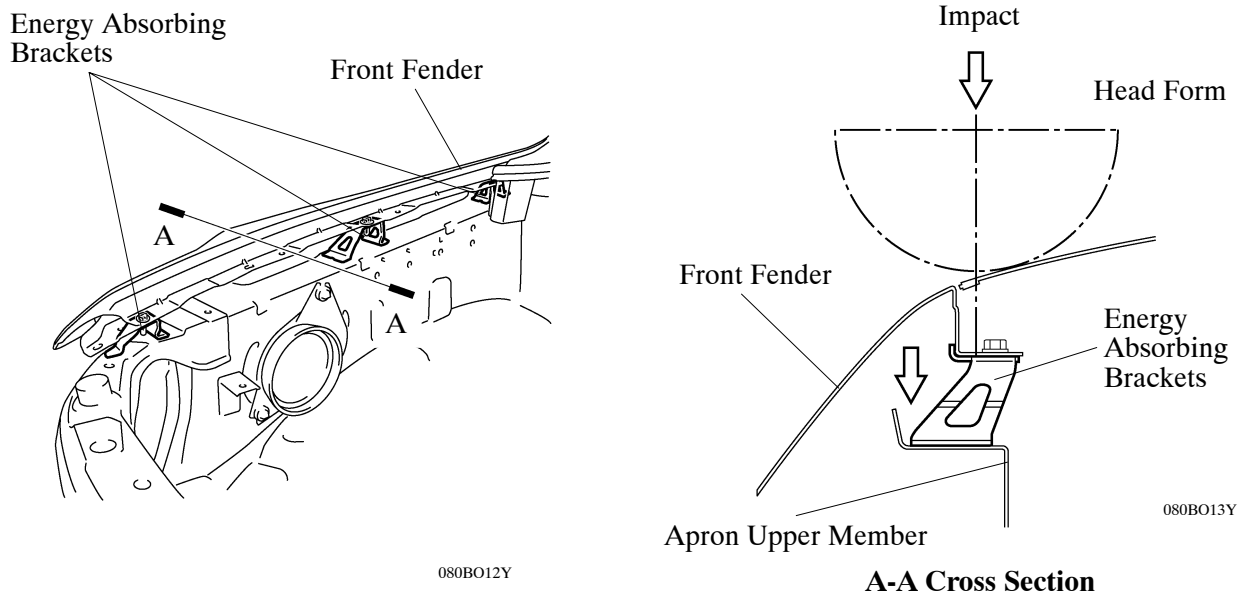


Impact absorption shape

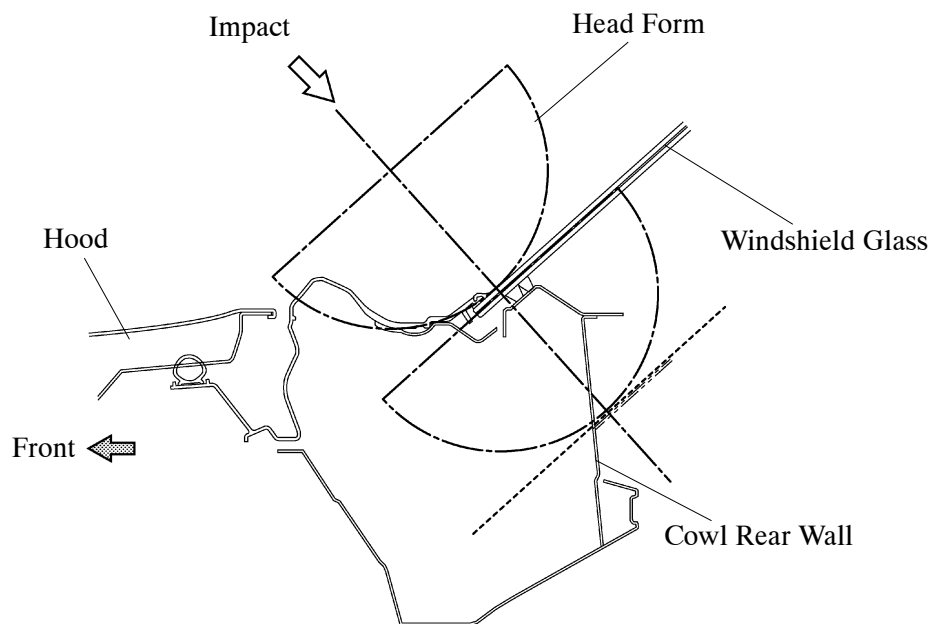
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A-A Cross Section

- Energy absorbing brackets are used in the joint portion of the front fender. Thus, a certain deformation stroke in the event of a head form collision has been ensured and help reduce the impact.



- Both body rigidity and energy absorption stroke have been achieved by optimizing the shape at the back of the cowl. Thus, good excellent pedestrian protection performance has been secured.



Cross Section at Lower Portion of Windshield